

PHIL 308 – Philosophy of Science

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Some Basics about Philosophical Arguments

What is an Argument?

In ordinary parlance, having an ‘argument’ often stands for having a dispute, and/or a disagreement that may even include fistfights. However, in philosophy we understand ‘Argument’ in a restrictive and somewhat technical fashion. *In Philosophy to give an argument is to make a claim and offer other claims as reasons for its acceptance.*

Here is a handy **definition** of Argument:

An Argument is A COLLECTION OF CLAIMS, one of which is meant to be supported by the others.

What is a claim?

A claim is a statement that is put forth *as being true*. For example, ‘London is pretty’.

Example of an argument:

If your car starts then your battery is alive.

Your car starts.

Hence your battery is alive.

Constituents of an argument:

Notice in our above example, the claim ‘(Hence) your battery is alive’ is what we are furnishing reasons to believe for, that is, arguing for. What we are trying to get others to believe or accept is called the CONCLUSION of the argument.

The reasons provided are called the PREMISES.

Here is another argument:

Premise: She’s armed.

Conclusion: She’s dangerous.

Difference between good arguments and bad arguments:

A **good argument** (in philosophical terms, a *sound* argument) is one *where the proposed support for a claim (i.e., the premise(s)) does support the claim (i.e., the conclusion)*. This support can be of various degrees depending on the nature of an argument. However, for now, we will ignore these complications. A good argument will also be such that *its premises are true*. Consider the following example:

Premise 1: If the temperature is above 55 degrees then one can safely skate in our lake.

Premise 2: The temperature is 60 degrees today.

Therefore, You can skate today.

In this argument though the premises provide sufficient support for the conclusion (since the premises cannot be true without the conclusion also being true), still it falls short of being a good argument. The reason is, one of its premises, namely, the first one is false. If the temperature is above 55 degrees then one’s safe skating is not guaranteed in any lake!

Exercises in Argument Diagramming:

Nothing is demonstrable unless the contrary implies a contradiction. Nothing that is distinctly conceivable implies a contradiction. Whatever we conceive as existent, we can also conceive as non-existent. There is no being, therefore, whose non-existence implies a contradiction. Consequently there is no being whose existence is demonstrable.

Nothing in the world—or out of it!—can possibly be conceived that could be called ‘good’ without qualification except a GOOD WILL. Mental talents such as intelligence, wit, and judgment, and temperaments such as courage, resoluteness, and perseverance are doubtless in many ways good and desirable; but they can become extremely bad and harmful if the person’s character isn’t good—i.e. if the will that is to make use of these gifts of nature isn’t good. Similarly with gifts of fortune. Power, riches, honour, even health, and the over-all well-being and contentment with one’s condition that we call ‘happiness’, create pride, often leading to arrogance, if there isn’t a good will to correct their influence on the mind. . . . Not to mention the fact that the sight of someone who shows no sign of a pure and good will and yet enjoys uninterrupted prosperity will never give pleasure to an impartial rational observer. So it seems that without good will one can’t even be worthy of being happy.

Two main *kinds* of arguments: Deductive OR Inductive

In a **Deductive argument**, the premise(s) offer *absolute certainty* for the conclusion. In a good deductive argument, if the premises are true, then the conclusion has to be true.

In an **Inductive argument**, the premise(s) offer *likelihood* for the conclusion. In a good Inductive argument, if the premises are true, then the conclusion is highly likely.

Exercises on the Distinction between DEDUCTIVE and INDUCTIVE Arguments:

All elements with atomic weights greater than 64 are metals. Z is an element with an atomic weight of 79. Therefore, Z is a metal.

Deductive OR Inductive?

How can you tell?

What factors will you consult to determine whether the argument is good or bad?

Some antibiotics are effective for treating certain bacterial infections. You have a bacterial infection. You are taking antibiotic Q. Thus the antibiotic you are taking will be effective in treating your bacterial infection.

Deductive OR Inductive?

How can you tell?

What factors will you consult to determine whether the argument is good or bad?